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Heat management system for vehicle

Abstract

When front and rear air conditioning units perform dehumidification heating operation, and when a battery is to be cooled, a refrigeration cycle circuit performs cooling operation and supplies a liquid-phase refrigerant to front and rear evaporators and a battery cooling heat exchanger. A heating circuit supplies a high temperature liquid heated by an electric heater to front and rear heater cores. Air cooled by the front and rear evaporators is warmed by the front and rear heater cores and supplied to a passenger compartment. A battery cooling liquid cooled by the battery cooling heat exchanger is delivered to the battery and cools the battery.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-169430 filed on Oct. 21, 2022, which is incorporated herein by reference in its entirety including the specification, claims, drawings, and abstract.

TECHNICAL FIELD

(2) The present disclosure relates to a heat management system for air conditioning a passenger compartment of a vehicle and managing temperature of vehicle-mounted equipment.

BACKGROUND

(3) JP 2018-103643 A discloses a vehicle that includes a front air conditioning unit (2) for delivering air having adjusted temperature and humidity to a front side space of a passenger compartment and a rear air conditioning unit (3) for delivering air having adjusted temperature and humidity to a rear side space of the passenger compartment. The front and rear air conditioning units (2, 3) respectively include a heater core (27, 48) and an evaporator (26, 47), enabling air conditioning of the passenger compartment independently for the front side space and the rear side space.

(4) There is also known an air conditioner for a vehicle, the air conditioner being configured to perform a dehumidification heating operation in which air is heated using a heater core while being dehumidified as it is cooled using an evaporator in a predetermined region that is determined based on a relationship between outside air temperature and temperature of air blown from an air conditioning unit.

(5) It should be noted that reference numerals in the parentheses above are reference numerals used in JP 2018-103643 A, which are unrelated to reference numerals used herein.

(6) In a structure wherein a refrigeration cycle circuit for an air conditioning unit is used for cooling a battery that supplies power to a motor that drives a vehicle, sufficient air conditioning and sufficient battery cooling sometimes cannot be performed when a refrigerant is supplied to front and rear air conditioning units and the refrigerant is further supplied for cooling the battery. During, in particular, the dehumidification heating operation of the air conditioner, the refrigerant that is supplied to the evaporators of the front and rear air conditioning units and to a battery cooling heat exchanger may be in a gas-liquid two-phase state. This causes a possibility that the flow rate of the refrigerant that is supplied to the evaporators and to the battery cooling heat exchanger cannot be controlled sufficiently, resulting in failure to achieve demanded air conditioning performance and battery cooling performance.

SUMMARY

(7) The present disclosure provides a vehicle heat management system that enables demanded air conditioning performance and battery cooling performance in a dehumidification heating operation region.

(8) According to an aspect of the present disclosure, there is provided a heat management system for a vehicle, the heat management system comprising a front air conditioning unit configured to perform air conditioning of a front side space of a passenger compartment of the vehicle; a rear air conditioning unit configured to perform air conditioning of a rear side space of the passenger compartment; a battery cooling circuit configured to circulate a cooling liquid for cooling a battery from which power is supplied to an electric motor for driving the vehicle; a refrigeration cycle circuit configured to supply a refrigerant to the front air conditioning unit and the rear air conditioning unit and configured to supply the refrigerant to a battery cooling heat exchanger for exchanging heat between the cooling liquid of the battery cooling circuit and the refrigerant; and a heating circuit configured to produce a high temperature liquid using a heat source or the refrigerant of the refrigeration cycle circuit and configured to supply the high temperature liquid to the front air conditioning unit and the rear air conditioning unit. A dehumidification heating operation region is defined based on an outside air temperature and a demanded blowing air temperature representing a target value to which a temperature of air fed from the front air conditioning unit and the rear air conditioning unit is controlled. When the front air conditioning unit and the rear air conditioning unit are operating in the dehumidification heating operation region, and when a demand to cool the battery is greater than or equal to a first demand value, the refrigeration cycle circuit performs cooling operation to supply the refrigerant to the front air conditioning unit, the rear air conditioning unit, and the battery cooling heat exchanger. Also, the heating circuit produces the high temperature liquid using the heat source. Further, the battery cooling circuit cools the battery using the cooling liquid cooled by the refrigerant of the refrigeration cycle circuit through the battery cooling heat exchanger.

(9) The refrigeration cycle circuit performs cooling operation, thereby turning the refrigerant into a liquid phase only. This enables delivery of the liquid-phase refrigerant to the front air conditioning unit, the rear air conditioning unit, and the battery cooling heat exchanger.

(10) In the above-described heat management system for a vehicle, when the demand to cool the battery is greater than or equal to a second demand value that is higher than the first demand value, and when an amount of electricity stored in the battery is less than a predetermined value, supply of the refrigerant to the front air conditioning unit and the rear air conditioning unit by the

refrigeration cycle circuit may be suspended. A reduction in power consumed by the refrigeration cycle circuit enables a longer maximum driving range achievable by the electric motor.

(11) In the above-described heat management system for a vehicle, the refrigeration cycle circuit may include a compressor configured to compress the refrigerant and an exterior heat exchanger configured to exchange heat with outside air, enabling, during the cooling operation, the entire amount of the refrigerant that has been compressed by the compressor and discharged to pass through the exterior heat exchanger and turn into a liquid.

(12) In the above-described heat management system for a vehicle, the heat source of the heating circuit may comprise one or both of an engine for driving the vehicle and an electric heater.

(13) Under circumstances where the front air conditioning unit and the rear air conditioning unit are both operating in the dehumidification heating operation region, and where the battery should be cooled, the refrigerant that has turned into a liquid phase can be delivered to the front and rear air conditioning units and the battery cooling heat exchanger. The amounts of refrigerant delivered to the front air conditioning unit, the rear air conditioning unit, and the battery cooling heat exchanger can be adjusted more appropriately than with a structure in which a gas-liquid two-phase refrigerant is delivered.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Embodiments of the present disclosure will be described based on the following figures, wherein:

(2) FIG. 1 illustrates a schematic structure of a vehicle that incorporates a heat management system according to an embodiment of the present disclosure;

(3) FIG. 2 illustrates a front air conditioning unit and its controller;

(4) FIG. 3 schematically illustrates a structure of a heat management system according to the embodiment;

(5) FIG. 4 illustrates conditions that define operation modes of an air conditioner;

(6) FIG. 5 illustrates an operating state of the heat management system and, in particular, illustrates a state in which a heating operation is performed using an engine cooling liquid;

(7) FIG. 6 illustrates an operating state of the heat management system and, in particular, illustrates a state in which a heating operation is performed using an electric heater;

(8) FIG. 7 illustrates an operating state of the heat management system and, in particular, illustrates a state in which a heating operation is performed using a heat pump operation of a refrigeration cycle circuit;

(9) FIG. 8 illustrates an operating state of the heat management system and, in particular, illustrates a state in which a cooling operation is performed;

(10) FIG. 9 illustrates an operating state of the heat management system and, in particular, illustrates a state in which a parallel dehumidification heating operation is performed;

(11) FIG. 10 illustrates an operating state of the heat management system and, in particular, illustrates a state in which a serial dehumidification heating operation is performed;

(12) FIG. 11 illustrates an operating state of the heat management system and, in particular, illustrates a state in which front and rear air conditioning units are operating with a battery being cooled in a dehumidification heating region;

(13) FIG. 12 illustrates an operating state of the heat management system and, in particular, illustrates a state in which the demand to cool the battery is high with the amount of stored electricity in the battery being low in a dehumidification heating region; and

(14) FIG. 13 illustrates part of a control flow of the heat management system.

DESCRIPTION OF EMBODIMENTS

(15) An embodiment of the present disclosure will be described below with reference to the accompanying drawings. FIG. 1 is a schematic diagram illustrating a schematic structure of a heat management system 12 of a vehicle 10. As a prime motor for driving the vehicle 10, the vehicle 10 incorporates an engine 14 and two electric motors 16F and 16R that drive front wheels and rear wheels, respectively. The vehicle 10 incorporates a battery 18 that supplies power to the electric motors 16F and 16R. During braking, the battery 18 is charged with power generated by the electric motors 16F and 16R. The vehicle may be a vehicle with a single electric motor that drives only either front wheels or rear wheels. Further, the vehicle may be a vehicle with no engine and with one or more electric motors that drive either front wheels or rear wheels or both. For simplicity, the electric motors 16F and 16R are hereinafter simply referred to as a motor 16.

(16) The heat management system 12 performs cooling of the engine 14, the motor 16, and the battery 18 and further performs air conditioning of a passenger compartment 20. The engine 14 has a cooling system that includes an engine radiator 22 that dissipates heat generated by the engine, using an engine cooling liquid. The engine cooling liquid flows through a pipe that connects between the engine 14 and the engine radiator 22, circulating between the engine 14 and the engine radiator 22. In FIG. 1, the pipe that connects between the engine 14 and the engine radiator 22 is not illustrated. The motor 16 has a cooling system that includes a motor radiator 24 that dissipates heat generated by the motor 16, using a motor cooling liquid. The motor cooling liquid flows through a pipe that connects between the motor 16 and the motor radiator 24, circulating between the motor 16 and the motor radiator 24. In FIG. 1, the pipe that connects between the motor 16 and the motor radiator 24 is not illustrated.

(17) The heat management system 12 includes an air conditioner 26 that performs air conditioning of the passenger compartment 20. The air conditioner 26 has an air conditioning unit 28 that supplies air having adjusted temperature, humidity, or the like to the passenger compartment 20. The air conditioning unit 28 includes a front air conditioning unit 28F that performs air conditioning of a front seat side space of the passenger compartment 20, and a rear air conditioning unit 28R that performs air conditioning of a rear seat side space of the passenger compartment 20. The air conditioner 26 includes a refrigeration cycle circuit 30 that delivers a refrigerant to the air conditioning unit 28, and a heating circuit 32 that delivers a heated liquid to the air conditioning unit 28. The refrigeration cycle circuit 30 includes a compressor 34 that compresses a refrigerant and an exterior condenser 36 that cools the refrigerant compressed by the compressor 34 using outside air, thereby converting the refrigerant into a liquid. The compressor 34 may be an electrically powered compressor that is driven by an electric motor, and the output of the compressor 34 can be adjusted by controlling the rotational speed of the electric motor. The heating circuit 32 includes an electric heater 38 as a heat source. The refrigeration cycle circuit 30 and the heating circuit 32, including other cooling systems of the heat management system 12, will be later described in more detail.

(18) The heat management system 12 further includes a battery cooling circuit 40 that cools the battery 18. The battery cooling circuit 40 supplies, to the battery 18, a battery cooling liquid that is cooled by the refrigerant of the refrigeration cycle circuit 30 in a battery cooling heat exchanger 41, thereby cooling the battery 18.

(19) FIG. 2 schematically illustrates a structure of the front air conditioning unit 28F. The front air conditioning unit 28F has a front evaporator 42F constituting one of the components of the refrigeration cycle circuit 30, and a front heater core 44F constituting one of the components of the heating circuit 32, and further has a unit case 46 that houses the front evaporator 42F and the front heater core 44F. The unit case 46 has an air inlet 48 for introducing air into the unit case 46. The air inlet 48 includes an inside air inlet 48C for introducing air from inside the passenger compartment 20, and an outside air inlet 48E for introducing air from outside the vehicle. The unit case 46 has an air outlet 50 for feeding air-conditioned air toward a predetermined area in the passenger compartment 20. The air outlet 50 includes a head outlet 50H for feeding airflow toward the head

and its vicinity of a passenger who is seated in a front seat, a foot outlet **50F** for feeding airflow toward the feet of the front seat passenger, and a defroster outlet **50D** for feeding airflow toward the interior surface of a windshield. The airflow fed from the head outlet **50H** passes through a duct in an instrument panel (not illustrated) and blows toward the head and its vicinity of the front seat passenger in the passenger compartment **20** via multiple air vents formed in the instrument panel. The airflow fed from the foot outlet **50F** blows toward the feet of the front seat passenger either directly from the foot outlet **50F** or after passing through a duct in the instrument panel. The airflow fed from the defroster outlet **50D** passes through a duct in the instrument panel and blows via an air vent provided facing the vicinity of the lower edge of the windshield. The front air conditioning unit **28F** has an air blower **52** that is disposed in a portion upstream of the unit case **46**. The air blower **52** produces airflow from the air inlet **48** toward the air outlet **50**.

(20) An inside/outside air switching door **54** is disposed in a portion where a flow channel from the inside air inlet **48C** meets a flow channel from the outside air inlet **48E**. The inside/outside air switching door **54** is pivotable between a position in which the inside air inlet **48C** is closed and a position in which the outside air inlet **48E** is closed, so that the mixing ratio of inside air and outside air is adjusted in accordance with the angle of pivot. An air mixing door **56** is disposed between the front evaporator **42F** and the front heater core **44F**. The air mixing door **56** pivots, thereby adjusting the amount of air passing through the front heater core **44F** after having passed through the front evaporator **42F**. For each air outlet **50**, a corresponding outlet door **58** for opening and closing the air outlet **50** is provided. Specifically, a head outlet door **58H** is provided for the head outlet **50H**, a foot outlet door **58F** is provided for the foot outlet **50F**, and a defroster outlet door **58D** is provided for the defroster outlet **50D**, so that the amount of air blown from each outlet **50** is adjusted in accordance with the open amount of the corresponding outlet door **58**.

(21) The amount of air blown by the air blower **52**, the angles of pivot of the inside/outside air switching door **54** and the air mixing door **56**, and the open amount of each outlet door **58** are controlled by a controller **60**. Based on conditions set by a passenger and environment conditions, the controller **60** controls the angles of pivot of the inside/outside air switching door **54** and the air mixing door **56**, the open amount of each outlet door **58**, and the amount of air blown by the air blower **52**. The passenger sets a desired temperature using a temperature setting switch **62** and further sets an air outlet **50** through which he/she wishes to have air blown, using an air vent switching switch **64**. For selection of the air outlet **50**, there can be selected a blow mode such as a blow mode in which one of the air outlets **50H**, **50F**, and **50D** is selected singly, or a blow mode in which air is blown from both of the head outlet **50H** and the foot outlet **50F**. The passenger can set an automatic air conditioning mode using an automatic air conditioning switch **66**, and in this case, the air outlet **50** through which the controller **60** causes air to be blown in accordance with a desired temperature and environment conditions is selected according to a predetermined program. An air temperature of the passenger compartment **20** detected by a room temperature sensor **68**, an outside air temperature detected by an outside air temperature sensor **70**, a liquid temperature of the heating circuit **32** detected by a liquid temperature sensor **72**, an amount of solar radiation entering the passenger compartment **20** detected by a solar radiation sensor **74**, and the like are input to the controller **60**. Further, a temperature of air that has just passed through the front evaporator **42F** (evaporator outlet temperature) detected by an evaporator outlet temperature sensor **76** disposed immediately after the front evaporator **42F** is input to the controller **60**.

(22) The rear air conditioning unit **28R** has a structure that is almost the same as that of the front air conditioning unit **28F**, and an illustration corresponding to FIG. 2 is not provided. The rear air conditioning unit **28R** has a unit case that houses a rear evaporator **42R** and a rear heater core **44R** (see FIG. 1), and the unit case has an air inlet and an air outlet. In the rear air conditioning unit **28R**, the air inlet may include no outside air inlet, and the air outlet may include no defroster outlet. As with the front air conditioning unit **28F**, the rear air conditioning unit **28R** can deliver air-conditioned air either toward the head and its vicinity of a passenger in a rear seat or toward the

feet of the rear seat passenger, or both.

(23) The rear air conditioning unit **28R** can be switched on or off by a passenger. In response to operation of a rear air conditioning switch **67** by a passenger, the controller **60** operates not only the front air conditioning unit **28F** but also the rear air conditioning unit **28R**. Temperature setting and blow mode selection can be performed independently for the front air conditioning unit **28F** and the rear air conditioning unit **28R**.

(24) The controller **60** is a processor that controls the air conditioner **26** according to a predetermined program, operating to achieve the temperature and air blow mode desired by the passenger in accordance with the above-described temperatures, amount of solar radiation, and the like.

(25) FIG. 3 schematically illustrates a structure of the heat management system **12**. The components already described are denoted by the same reference numerals. The refrigeration cycle circuit **30** includes not only the compressor **34**, the exterior condenser **36**, and the front and rear evaporators **42F** and **42R** described above, but also a liquid cooling condenser **78** that exchanges heat with the heating circuit **32**, and the battery cooling heat exchanger **41** that exchanges heat with the battery cooling circuit **40**. The refrigeration cycle circuit **30** has electric expansion valves **84** and **86** capable of adjusting the open amount respectively located upstream of the front evaporator **42F** and the battery cooling heat exchanger **41**, and has an expansion valve **88** and an electromagnetic valve **90** located upstream of the rear evaporator **42R**. The electric expansion valves **84** and **86** cannot be fully shut off, and even with the minimum open amount, a small amount of refrigerant is supplied to the front evaporator **42F** and the battery cooling heat exchanger **41**. On the other hand, supply of the refrigerant to the rear evaporator **42R** can be fully stopped when the electromagnetic valve **90** is closed. Further, a heating expansion valve **92** is provided upstream of the exterior condenser **36**. The heating expansion valve **92** is capable of adjusting the open amount and may be an electric expansion valve. During a heating operation of the refrigeration cycle circuit **30**, the refrigerant expands as it passes through the heating expansion valve **92** with a reduced open amount, absorbing heat when it vaporizes in the exterior condenser **36**. As such, during the heating operation, the exterior condenser **36** functions as an evaporator. During a cooling operation of the refrigeration cycle circuit **30**, the heating expansion valve **92** is fully open, simply letting the refrigerant pass. The capacity of the refrigeration cycle circuit **30** is adjusted by adjusting the output of the compressor **34** and adjusting the open amounts of the expansion valves **84**, **86**, **88**, and **92**.

(26) The refrigeration cycle circuit **30** has a first bypass flow channel **94** disposed in parallel with the front and rear evaporators **42F** and **42R** and the battery cooling heat exchanger **41**. The refrigerant can bypass the front and rear evaporators **42F** and **42R** and the battery cooling heat exchanger **41** when it passes through the first bypass flow channel **94**. The refrigeration cycle circuit **30** also has a second bypass flow channel **96** disposed in parallel with the exterior condenser **36**, and the refrigerant can bypass the exterior condenser **36** when it passes through the second bypass flow channel **96**.

(27) The battery cooling circuit **40** includes the battery **18**, the battery cooling heat exchanger **41**, and a battery cooling circuit pump **98** for circulating a cooling liquid between the battery **18** and the battery cooling heat exchanger **41**. The battery **18** is cooled as the cooling liquid cooled by the battery cooling heat exchanger **41** is delivered to the battery **18**. The battery **18** has a battery temperature sensor **100** for detecting the temperature of the battery **18** and an electricity storage amount sensor **102** for detecting the amount of electricity stored in the battery **18**. The level of demand to cool the battery **18** is determined based on, for example, the temperature of the battery **18**, and the battery cooling circuit **40** is controlled in accordance with the determined level. The electricity storage amount sensor **102** continuously monitors power that is input and output to and from the battery **18**, and input and output power values are added up to calculate a current amount of stored electricity.

(28) The heating circuit **32** includes the electric heater **38**, the front and rear heater cores **44F** and **44R**, the liquid cooling condenser **78**, and a heating circuit pump **104** that delivers a high temperature liquid. The heating circuit pump **104** causes a circulating liquid to circulate through the electric heater **38**, the front and rear heater cores **44F** and **44R**, and the liquid cooling condenser **78**. The liquid cooling condenser **78** heats the circulating liquid of the heating circuit **32** using a high temperature refrigerant that is compressed by the compressor **34** of the refrigeration cycle circuit **30**, thereby producing a high temperature liquid. The high temperature liquid, which is the circulating liquid heated either by the electric heater **38** or by the liquid cooling condenser **78**, is delivered to the front and rear heater cores **44F** and **44R**. The heating circuit **32** shares a heating/cooling liquid with an engine cooling circuit **106**, and the engine **14** can be used as a heat source. Whether the high temperature liquid delivered to the front and rear heater cores **44F** and **44R** is supplied from the side on which the engine **14** is located, or whether it is supplied from the side on which the electric heater **38** and the liquid cooling condenser **78** are located, is determined in response to operation of a three-way valve **108**.

(29) The engine cooling circuit **106** includes the engine **14** and the engine radiator **22**, and further includes an engine cooling circuit pump **110** for circulating an engine cooling liquid between the engine **14** and the engine radiator **22**. The engine cooling circuit **106** includes a radiator bypass flow channel **112** disposed in parallel with the engine radiator **22**, and the engine cooling liquid can be circulated while bypassing the engine radiator **22**. In a state in which the engine **14** is cold, such as while being warmed up, the engine cooling circuit **106** does not deliver the engine cooling liquid to the engine radiator **22** but circulates the engine cooling liquid through the radiator bypass flow channel **112**, thereby enabling a rapid increase in temperature of the engine cooling liquid. As described above, fluid for the heating circuit **32** can also serve as the engine cooling liquid.

(30) A flow channel through which a refrigerant or fluid flows in the refrigeration cycle circuit **30**, the heating circuit **32**, the battery cooling circuit **40**, and the engine cooling circuit **106** is changed in accordance with predetermined conditions. A change in refrigerant or fluid flow channel is achieved not only by the three-way valve **108** and the electromagnetic valve **90** as described above, but also in response to operation of multiple valves that are not illustrated but are provided as desired for the respective circuits. The opening/closing of these valves and their open amounts may be controlled by the controller **60**. In response to demands, the controller **60** also controls the output of the compressor **34**, the discharge flow rate of the battery cooling circuit pump **98**, and the discharge flow rate of the heating circuit pump **104**.

(31) The heat management system **12** operates in some operation modes that suit predetermined conditions. The predetermined conditions are determined based on, for example, the outside air temperature, the temperature of airflow blowing from the air conditioner **26** based on the passenger's demands (demanded blowing air temperature), and the demand to cool the battery **18** and the amount of electricity stored in the battery **18**.

(32) FIG. 4 illustrates an example of conditions that define the operation mode of the heat management system **12**, or, in particular, the operation mode of the air conditioner **26**. The heat management system **12** operates in a heating mode; that is, performs heating operation, in a heating region H in which the outside air temperature is less than a predetermined air temperature T1 (for example, 0° C.). The heat management system **12** operates in a cooling mode; that is, performs cooling operation, in a cooling region C in which the outside air temperature is higher than the predetermined air temperature T1, the demanded blowing air temperature is low, and the difference between the demanded blowing air temperature and the outside air temperature is large. In dehumidification heating regions Dp and Ds that are intermediate between the heating region H and the cooling region C, the heat management system **12** operates in a dehumidification heating mode in which air taken into the air conditioning unit **28** is first cooled for dehumidification and then heated to the demanded blowing air temperature; that is, performs dehumidification heating operation. The dehumidification heating region includes a parallel dehumidification heating region

Dp and a serial dehumidification heating region Ds. The heat management system **12** performs parallel dehumidification heating operation in the parallel dehumidification heating region Dp and performs serial dehumidification heating operation in the serial dehumidification heating region Ds. The parallel dehumidification heating operation is an operation mode used in a region in which heating is set higher than the serial dehumidification heating operation. For the front air conditioning unit **28F** and the rear air conditioning unit **28R** for which the demanded blowing air temperature can be set independently, the demanded blowing air temperature for a predetermined one of the air conditioning units **28**, such as for the front air conditioning unit **28F**, may be an element that defines the operation mode. Alternatively, the demanded blowing air temperature for the air conditioning unit **28** for which a lower temperature is set may be an element that defines the operation mode. The operation of the heat management system **12** in the respective operation modes will be described below.

(33) FIGS. **5** to **7** illustrate operating states of the heat management system **12** in the heating region H. In the following description, the front heater core **44F** and the rear heater core **44R** are collectively referred to as heater core **44** where particular distinction is unnecessary.

(34) FIG. **5** illustrates an operating state observed when the temperature of a cooling liquid in the engine **14** is sufficiently high. When the temperature of the engine cooling liquid is high, some of the engine cooling liquid circulated by the engine cooling circuit pump **110** is supplied to the heater core **44**. In response to operation by a passenger, the engine cooling liquid may be supplied to one or both of the front and rear heater cores **44F** and **44R**.

(35) FIG. **6** illustrates an operating state observed when the temperature of the engine cooling liquid is low. When the temperature of the engine cooling liquid is low, the electric heater **38** produces a high temperature liquid, and this high temperature liquid is supplied to the heater core **44** by the heating circuit pump **104**. In response to operation by a passenger, the high temperature liquid supplied from the electric heater **38** may be supplied to one or both of the front and rear heater cores **44F** and **44R**. For a vehicle that travels only by power from the electric motor **16** without the engine **14**, heating is performed either by the electric heater **38** or by heat pump operation of the refrigeration cycle circuit **30**, which will be described below.

(36) FIG. **7** illustrates an operating state in which heating is performed through heat pump operation of the refrigeration cycle circuit **30**. The refrigerant that has been compressed by the compressor **34** to have high temperature turns into a liquid in the liquid cooling condenser **78** as it is cooled by fluid circulating in the heating circuit **32**. In this process, the circulating liquid in the heating circuit **32** turns into a high temperature liquid as it is heated by the high temperature refrigerant. The refrigerant in the refrigeration cycle circuit **30**, which has turned into a liquid in the liquid cooling condenser **78**, expands through the heating expansion valve **92** and vaporizes in the exterior condenser **36**, thereby absorbing heat from outside air. That is, the exterior condenser **36** functions as an evaporator in this process. The refrigerant that has vaporized returns to the compressor **34** through the first bypass flow channel **94**. The high temperature liquid heated by the liquid cooling condenser **78** is supplied to the heater core **44** by the heating circuit pump **104**. In response to operation by a passenger, the high temperature liquid supplied from the electric heater **38** may be supplied to one or both of the front and rear heater cores **44F** and **44R**. During this heat pump operation, the passenger compartment **20** is heated using heat pumped up from outside air.

(37) FIG. **8** illustrates an operating state of the heat management system **12** in the cooling region C. In the following description, the front evaporator **42F** and the rear evaporator **42R** are collectively referred to as evaporator **42** where particular distinction is unnecessary. The refrigerant that has been compressed by the compressor **34** dissipates heat into outside air in the exterior condenser **36**, turning into a liquid as the refrigerant itself is cooled. The refrigerant that has turned into a liquid expands through the electric expansion valve **84** and the expansion valve **88** and respectively vaporizes in the front and rear evaporators **42F** and **42R**, thereby absorbing heat. As a result, the passenger compartment **20** is cooled. When it is unnecessary to cool the rear side of the passenger

compartment **20**, such as when there is no passenger in the rear seat, the electromagnetic valve **90** may be closed, thereby stopping supply of the refrigerant to the rear evaporator **42R**.

(38) FIG. **9** illustrates an operating state of the heat management system **12** in the parallel dehumidification heating region Dp. The refrigerant that has been compressed by the compressor **34** dissipates heat into the circulating liquid of the heating circuit **32** in the liquid cooling condenser **78**. As a result, the high temperature liquid of the heating circuit **32** is produced and supplied to the heater core **44**. As in the above-described heating operation, the high temperature liquid may be supplied to one or both of the front heater core **44F** and the rear heater core **44R**. During the parallel dehumidification heating operation, the amount of heat transferred by the refrigeration cycle circuit **30** is smaller than in the heating operation with the heat pump and the cooling operation, and thus the demand on the capacity of the refrigeration cycle circuit **30** is smaller. Therefore, the output of the compressor **34** is low, and the refrigerant that has dissipated heat in the liquid cooling condenser **78** does not fully turn into a liquid but remains in a gas-liquid two-phase state. Some of the refrigerant flows toward the exterior condenser **36** through the narrowed heating expansion valve **92**, and at least some of it vaporizes in the exterior condenser **36**, thereby absorbing heat. In this process, the exterior condenser **36** functions as an evaporator. The refrigerant that has passed through the exterior condenser **36** returns to the compressor **34** through the first bypass flow channel **94**. The remaining portion of the refrigerant that has passed through the liquid cooling condenser **78** flows toward the evaporator **42** through the second bypass flow channel **96**. In the evaporator **42**, at least some of the liquid-phase refrigerant vaporizes, thereby absorbing heat. After passing through the evaporator **42**, the refrigerant returns to the compressor **34**. When it is unnecessary to cool the rear side of the passenger compartment **20**, such as when there is no passenger in the rear seat, the electromagnetic valve **90** may be closed, thereby stopping supply of the refrigerant to the rear evaporator **42R**.

(39) In the air conditioning unit **28**, air is cooled through the evaporator **42**, thereby causing dehumidification as water vapor is cooled and condensed. As the cooled air is heated by the heater core **44**, the air conditioning unit **28** feeds warm dry air to the passenger compartment **20**.

(40) FIG. **10** illustrates an operating state of the heat management system **12** in the serial dehumidification heating region Ds. The refrigerant that has been compressed by the compressor **34** dissipates heat into the circulating liquid of the heating circuit **32** in the liquid cooling condenser **78**. As a result, the high temperature liquid of the heating circuit **32** is produced and supplied to the heater core **44**. As in the above-described heating operation, the high temperature liquid may be supplied to one or both of the front heater core **44F** and the rear heater core **44R**. The entire amount of the refrigerant that has passed through the liquid cooling condenser **78** is delivered to the exterior condenser **36**. As in the parallel dehumidification heating operation, the refrigerant absorbs heat in the exterior condenser **36** when the refrigerant dissipates a large amount of heat in the liquid cooling condenser **78**; that is, when heating is set high. As the open amount of the heating expansion valve **92** is narrowed, some of the refrigerant vaporizes in the exterior condenser **36**, thereby absorbing heat. On the other hand, when heating can be set low, the refrigerant dissipates heat in the exterior condenser **36**. In this case, the heating expansion valve **92** is set fully open. During the serial dehumidification heating operation, the amount of heat transferred by the refrigeration cycle circuit **30** is smaller than in the heating operation with the heat pump and the cooling operation, and thus the demand on the capacity of the refrigeration cycle circuit is smaller. Therefore, the output of the compressor **34** is low, and the refrigerant that has dissipated heat in the liquid cooling condenser **78** and the exterior condenser **36** does not fully turn into a liquid but remains in a gas-liquid two-phase state. The refrigerant that is in a gas-liquid two-phase state flows toward the evaporator **42**. In the evaporator **42**, the liquid-phase refrigerant vaporizes, thereby absorbing heat. After passing through the evaporator **42**, the refrigerant returns to the compressor **34**. When it is unnecessary to cool the rear side of the passenger compartment **20**, such as when there is no passenger in the rear seat, the electromagnetic valve **90** may be closed, thereby stopping

supply of the refrigerant to the rear evaporator **42R**.

(41) During the heating operation, the refrigeration cycle circuit **30** transfers outside heat to inside the passenger compartment **20**, thereby heating the passenger compartment **20**, and during the cooling operation, the refrigeration cycle circuit **30** transfers heat inside the passenger compartment **20** to outside, thereby performing cooling. During the dehumidification heating operation, the refrigeration cycle circuit **30** absorbs heat from the evaporator **42** and dissipates heat from the heater core **44** through the heating circuit **32**, thereby transferring heat in the passenger compartment **20**. An amount of heat corresponding to the difference between an amount of heat absorbed in the evaporator **42** and an amount of heat dissipated in the liquid cooling condenser **78** is absorbed or dissipated in the exterior condenser **36**.

(42) During the dehumidification heating operation, the amount of heat transfer is less than that in the heating operation and the cooling operation, and the output of the compressor **34** is also reduced. Therefore, the refrigerant that has dissipated heat does not fully turn into a liquid but turns into a gas-liquid two-phase state. The refrigerant that is in a gas-liquid two-phase state may be supplied to multiple targets, and this may result in an unbalanced distribution of the liquid-phase refrigerant. To cool the battery **18**, the liquid-phase refrigerant should be reliably supplied to the battery cooling heat exchanger **41**. In particular, the liquid-phase refrigerant may be supplied insufficiently when the refrigerant is supplied to both of the front evaporator **42F** and the rear evaporator **42R** and further supplied to the battery cooling heat exchanger **41**.

(43) In response to a demand to cool the battery **18** while, in the dehumidification heating regions D_p and D_s , both of the front and rear air conditioning units **28F** and **28R** are operating (in dual mode), the refrigeration cycle circuit **30** performs cooling operation in which the liquid-phase refrigerant is supplied to the evaporator **42**, and the high temperature liquid heated by the electric heater **38** is supplied to the heater core **44**. During the cooling operation of the refrigeration cycle circuit **30**, the compressor **34** operates at a high output so that the refrigerant is compressed sufficiently, and the refrigerant dissipates heat in the exterior condenser **36**, thereby fully turning into a liquid. As a result, a necessary amount of the liquid-phase refrigerant is supplied to the front and rear evaporators **42F** and **42R** and the battery cooling heat exchanger **41**.

(44) In response to an increased demand to further cool the battery **18**, such as a rise in temperature of the battery **18**, the output of the compressor **34** is also increased correspondingly. As such, the power for driving the compressor **34** is also increased, resulting in increased power consumption. The increased power consumption shortens the maximum driving range achievable by the electric motor **16**. In this heat management system **12**, when the demand to cool the battery **18** is increased, and when the amount of electricity stored in the battery **18** decreases to less than a predetermined value, such as 20%, the air conditioner **26** suspends dehumidification. Specifically, supply of the refrigerant to the front and rear evaporators **42F** and **42R** is suspended. As the output of the compressor **34** can be lowered due to the suspension of supply of the refrigerant to the evaporator **42**, the power consumption can be lower. Without dehumidification, as the temperature of air that has passed through the evaporator **42** does not drop, less heating by the heater core **44** can be sufficient. Therefore, as the power supplied to the electric heater **38** can also be reduced, the power consumption can be lower. This can extend the maximum driving range achievable by the electric motor **16**, as the reduction in amount of electricity stored in the battery **18** is retarded.

(45) FIG. **11** illustrates an operating state of the heat management system **12** in the dehumidification heating regions D_p and D_s , in which the air conditioner **26** is operating in dual mode, and the battery cooling circuit **40** is operating. In the refrigeration cycle circuit **30**, the refrigerant that has been compressed by the compressor **34** is delivered to the exterior condenser **36**, where the entire amount of it turns into a liquid as it dissipates heat. The liquid-phase refrigerant is supplied to the front and rear evaporators **42F** and **42R** and the battery cooling heat exchanger **41**. The refrigerant that has been delivered to the front and rear evaporators **42F** and **42R** vaporizes there, absorbing heat from air in the passenger compartment **20**. The refrigerant that has

been delivered to the battery cooling heat exchanger **41** vaporizes there, absorbing heat from the battery cooling liquid of the battery cooling circuit **40**. On the other hand, the battery cooling liquid is cooled, and the cooled battery cooling liquid is delivered to the battery **18** and cools the battery **18**. In the heating circuit **32**, the electric heater **38** operates to produce the high temperature liquid, and the high temperature liquid is supplied to the heater core **44**. If the temperature of the cooling liquid in the engine **14** is sufficiently high, the engine cooling liquid may be supplied to the front and rear heater cores **44F** and **44R**.

(46) FIG. **12** illustrates an operating state of the heat management system **12** observed when the necessity for further cooling the battery **18** arises. In contrast to the operating state illustrated in FIG. **11**, supply of the refrigerant to the front and rear evaporators **42F** and **42R** is suspended. The temperature of the refrigerant that has been compressed by the compressor **34** rises, thereby producing the high temperature liquid of the heating circuit **32**. As the air conditioning unit **28** does not perform dehumidification, the temperature of air that has passed through the evaporator **42** does not drop, which involves being either not reheated or only slightly reheated by the heater core **44**. When the heating using the refrigerant of the refrigeration cycle circuit **30** is insufficient, the electric heater **38** also heats the circulating liquid of the heating circuit **32**.

(47) FIG. **13** illustrates a control flow of the heat management system **12** in which the front and rear air conditioning units **28F** and **28R** both perform dehumidification heating operation, and in which there is a battery cooling demand. According to this control flow, the controller **60** controls the heat management system **12**.

(48) When the rear air conditioning switch **67** is on; that is, in dual mode in which the front and rear air conditioning units **28F** and **28R** are both operating (**S100**), and when the operation conditions are in the dehumidification heating region **Dp** or **Ds** (**S102**), the controller **60** determines whether or not there is a demand for cooling the battery (**S104** and **S106**). When the battery cooling demand is greater than or equal to a first demand value, the controller **60** determines that the battery should be cooled using the battery cooling circuit **40** (**S104**). The battery cooling demand is determined based on, for example, the temperature of the battery **18** detected by the battery temperature sensor **100**. When, in step **S104**, the battery cooling demand is less than the first demand value, the process returns to the start of this control flow. When the battery cooling demand is greater than or equal to the first demand value, the controller **60** further determines whether the battery cooling demand is greater than or equal to a second demand value (**S106**). The second demand value corresponds to a demand to cool the battery further than the first demand value and corresponds to, for example, a further rise in battery temperature. When, in step **S106**, the battery cooling demand is less than the second demand value, then as illustrated in FIG. **11**, the controller **60** controls the refrigeration cycle circuit **30** to perform cooling operation and controls the heating circuit **32** to perform heating operation (**S108**). The controller **60** controls the refrigeration cycle circuit **30** to deliver the refrigerant to the battery cooling heat exchanger **41** of the battery cooling circuit **40**, and the controller **60** further controls the battery cooling circuit **40** to deliver the cooling liquid cooled by this refrigerant to the battery **18** (**S108**).

(49) Under circumstances where the refrigerant is supplied to three targets; that is, the front and rear evaporators **42F** and **42R** and the battery cooling heat exchanger **41**, the cooling operation of the refrigeration cycle circuit **30** enables delivery of only liquid-phase refrigerant to the three targets. This enables more reliable supply of the refrigerant to three targets than with a structure in which a gas-liquid two-phase refrigerant is delivered.

(50) When, in step **S106**, the battery cooling demand is greater than or equal to the second demand value, the controller **60** further determines whether the amount of electricity stored in the battery **18** is less than a predetermined value, for example, less than 20% (**S110**). When the amount of stored electricity is greater than or equal to the predetermined value, then the process proceeds to step **S108**. When the amount of stored electricity is less than the predetermined value, then the controller **60** suspends dehumidification of the air conditioner **26** (**S112**). Specifically, as illustrated

in FIG. 12, the controller **60** controls the refrigeration cycle circuit **30** to suspend supply of the refrigerant to the front and rear evaporators **42F** and **42R**. The refrigeration cycle circuit **30** supplies the refrigerant only to the battery cooling heat exchanger **41**. The heating circuit **32** may supply the high temperature liquid to the front and rear heater cores **44F** and **44R** as needed. (51) The suspension of dehumidification enables reduced increase in the output of the compressor **34** and also enables reduced reheating of air in the air conditioning unit **28**. This enables achievement of a longer maximum driving range by the electric motor **16** with reduced power consumption of the battery **18**.

REFERENCE SIGNS LIST

(52) **10** vehicle **12** heat management system **14** engine **16** electric motor **18** battery **20** passenger compartment **22** engine radiator **26** air conditioner **28** air conditioning unit **28F** front air conditioning unit **28R** rear air conditioning unit **30** refrigeration cycle circuit **32** heating circuit **34** compressor **36** exterior condenser **38** electric heater **40** battery cooling circuit **41** battery cooling heat exchanger **42** evaporator **42F** front evaporator **42R** rear evaporator **44** heater core **44F** front heater core **44R** rear heater core **60** controller **78** liquid cooling condenser **84**, **86** electric expansion valve **88** expansion valve **90** electromagnetic valve **92** heating expansion valve **94** first bypass flow channel **96** second bypass flow channel **98** battery cooling circuit pump **100** battery temperature sensor **102** electricity storage amount sensor **104** heating circuit pump **106** engine cooling circuit **108** three-way valve

Claims

1. A heat management system for a vehicle, the heat management system comprising: a front air conditioning unit configured to perform air conditioning of a front side space of a passenger compartment of the vehicle; a rear air conditioning unit configured to perform air conditioning of a rear side space of the passenger compartment; a battery cooling circuit configured to circulate a cooling liquid for cooling a battery from which power is supplied to an electric motor for driving the vehicle; a refrigeration cycle circuit configured to supply a refrigerant to the front air conditioning unit and the rear air conditioning unit and configured to supply the refrigerant to a battery cooling heat exchanger for exchanging heat between the cooling liquid of the battery cooling circuit and the refrigerant; and a heating circuit configured to produce a high temperature liquid using a heat source or the refrigerant of the refrigeration cycle circuit and configured to supply the high temperature liquid to the front air conditioning unit and the rear air conditioning unit, wherein, when the front air conditioning unit and the rear air conditioning unit are operating in a dehumidification heating operation region determined based on an outside air temperature and a demanded blowing air temperature representing a target value to which a temperature of air fed from the front air conditioning unit and the rear air conditioning unit is controlled, and when a demand to cool the battery is greater than or equal to a first demand value, the refrigeration cycle circuit performs cooling operation to supply the refrigerant to the front air conditioning unit, the rear air conditioning unit, and the battery cooling heat exchanger, the heating circuit produces the high temperature liquid using the heat source, and the battery cooling circuit cools the battery using the cooling liquid cooled by the refrigerant of the refrigeration cycle circuit through the battery cooling heat exchanger, wherein, when the demand to cool the battery is greater than or equal to a second demand value that is higher than the first demand value, and when an amount of electricity stored in the battery is less than a predetermined value, supply of the refrigerant to the front air conditioning unit and the rear air conditioning unit by the refrigeration cycle circuit is suspended.
2. The heat management system for a vehicle according to claim 1, wherein the refrigeration cycle circuit includes a compressor configured to compress the refrigerant and an exterior heat exchanger configured to exchange heat with outside air, thereby enabling, during the cooling operation, the total amount of the refrigerant that has been compressed by the compressor and discharged to pass

through the exterior heat exchanger and turn into a liquid.

3. The heat management system for a vehicle according to claim 1, wherein the heat source of the heating circuit comprises one or both of an engine for driving the vehicle and an electric heater.
